🌍 Final Project – From Simulation to Machine Learning

🎯 Objective

In this final project, you will complete the **entire pipeline** from detailed EnergyPlus simulation to data-driven prediction.
The goal is not only to build accurate models but also to **understand how physical simulation and machine learning differ** — in data source, scale, precision, and interpretation.

🧱 Step 1 – Generate Simulation Data

Use your simplified EnergyPlus model (`model_week4.idf`) to perform a **multi-parameter batch simulation**.

Parameter	Description	Levels
Location	Climate zones representing U.S. cities	San Francisco, Sacramento, Chicago, New York
Wall R-value	Wall insulation level (ft²·°F·hr/Btu)	14.3, 21.9, 29.7
WWR	Window-to-Wall Ratio	0.25, 0.40, 0.60
SHGC	Solar Heat Gain Coefficient	0.25, 0.40, 0.60
Occupancy Density	People per floor area	0.04, 0.06, 0.08
Lighting Density	Lighting power density (W/m²)	7.5, 9
Cooling Setpoint	Indoor cooling temperature (°C)	24, 26, 28

> 🔢 **Total simulation cases:** 4 × 3 × 3 × 3 × 3 × 2 × 3 = 1944 (or more if you add optional parameters)

Save the outputs:

- `results.csv` → cooling energy results
- `all_parameters.csv` → corresponding parameter combinations

⚙️ Step 2 – Feature Engineering

In this stage, you will **define and construct your own features** that best describe the relationship between building configuration and energy performance.

You may consider — but are not limited to — the following feature types:

Category	Possible Examples	Description
Climate Indicators	Cooling/Heating Degree Days (CDD/HDD), average or extreme temperatures, solar radiation	Capture how external conditions drive cooling demand
Design Variables	R-value, WWR, SHGC, setpoint, lighting/occupancy density	Represent envelope and operation parameters
Derived Features	Ratios, interaction terms (e.g., WWR × SHGC), normalized indicators	Capture nonlinear or combined effects

> 💡 There is **no single correct method** — you are encouraged to explore different feature definitions, transformations, or normalization strategies that improve model performance or interpretability.

> ✅ **Goal:** create a structured dataset linking simulation parameters and climate factors to building energy response.

🤖 Step 3 – Train Machine Learning Models

Using your combined dataset:

1. Define input (`X`) and target (`y`) variables.
2. Split your data into training (San Francisco, NYC, Chicago) and testing (Sacramento) sets.

3. Train and compare multiple regression models (including at least Linear, Decision Tree, Random Forest).
4. Evaluate performance using MSE, MAE, and R^2 metrics.

> ✅ **Goal:** test how well data-driven models can approximate simulation-based results.

🔍 Step 4 – Compare and Reflect

After completing your models, analyze and discuss the **differences between simulation and machine learning**.
Your reflection should go beyond numeric accuracy — focus on conceptual distinctions:

Aspect	Simulation (EnergyPlus)	Machine Learning
Data Source	Physics-based, derived from model geometry and materials	Pattern-based, relies on data relationships
Scale	Hourly/sub-hourly, spatially detailed	Aggregated across samples and features
Precision	Controlled by physical fidelity	Limited by data quality and coverage
Interpretability	Physically explicit	Statistical or black-box, depends on algorithm
Computation	Intensive, per simulation run	Lightweight after training
Use Case	Design analysis, calibration, validation	Prediction, optimization, surrogate modeling

> 🧠 **Reflection Question:**
> How do the **data scale** and **precision** differ between simulation and ML?
> What trade-offs arise when replacing physics-based detail with data-driven learning?

📝 Deliverables (required — slides only)

Submit a short slide deck (PDF or PPTX, 4–8 slides). The deck must be self-contained and clearly communicate your workflow, results, and interpretation. Required slide content:

1. Title slide — project title, authors, date (1 slide).
2. Climate distributions — 1 slide with plots showing outdoor temperature distributions for each location (histogram/violin + short summary stats).
3. ML evaluation — 1–2 slides that include:
   - Key metrics (MSE, RMSE, MAE, R^2) for the models you report,
   - Predicted vs actual scatter (with 1:1 line) and a brief note on major errors,
   - One compact feature-importance or interpretability plot.
4. Reflection & conclusions — 1–2 slides that:
   - Compare data-driven (ML) results vs physics-based simulation (strengths, limitations, recommended use cases),
   - State key takeaways and 1–2 recommended next steps.

📤 Submission instructions

- Filename: final_slides.pdf or final_slides.pptx
- Submit slides only; including data/notebooks/code is optional but encouraged (provide a link or archive if available).

Grading (slides-focused, 100 pts)

- Clarity & communication: 25 pts
- Climate & data visualisation: 25 pts
- ML evaluation (metrics + plots): 20 pts
- Reflection & interpretation: 30 pts

Submission Deadline

Submit your final slide deck (final_slides.pdf or final_slides.pptx) via the course LMS by 28/11/2025, 23:59 local time.
If you include links to data/code/notebooks, add them in an appendix slide and ensure the links are accessible.